

Derivative rules

$$\begin{aligned}\frac{d}{dx}(f(x)g(x)) &= \frac{df(x)}{dx}g(x) + f(x)\frac{dg(x)}{dx} \\ \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) &= \frac{1}{g^2(x)}\left[\frac{df(x)}{dx}g(x) - f(x)\frac{dg(x)}{dx}\right] \\ \frac{d}{dx}(f(g(x))) &= \frac{df(x)}{dx}\bigg|_{x=g(x)}\frac{dg(x)}{dx}\end{aligned}$$

$$\begin{aligned}\frac{d}{dx}(n^x) &= \ln(n)n^x, & \frac{d}{dx}\sin(x) &= \cos(x) \\ \frac{d}{dx}\cos(x) &= -\sin(x), & \frac{d}{dx}\log_n(x) &= \frac{1}{\ln(n)x}\end{aligned}$$